Control Flow Statements

Fundamentals of Programming – CSC100AB

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Contents

- Branch statements.
- Loop statements.

Contents

- **Branch statements.**
- Loop statements.

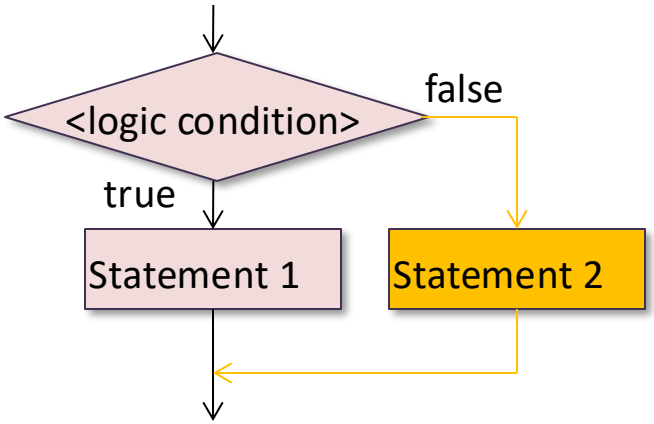
Branch statements

- if-else statement:
  - Syntax:

```
if (<logic condition>)
        <Statement 1>;
 [else
        <Statement 2>;]
```

Pseudo-code:

```
If <logic condition>
        Statement 1
 [Else
        Statement 2]
```



  - Examples:

```
// Full if-else
if (n > 0)
        a = a * 2;
else
        a = a / 2;
```

```
// No else
if (n > 0)
        a = a * 2;
```

```
// Compound statement
if (n > 0)
{
        a = a * 2;
        b = b + 1;
}
```

Branch statements

- if-else statement:
  - Notes:

    Logic condition encloses ( ).

    ➔ Value 1: true.

    ➔ Value 0: false.

    if-else compound statement.

    ➔ No **;** after **if** or **else**.

    if-else can be nested.

    ➔ **else** goes with nearest **if**.

```
if  n > 0                // Wrong
        a = a * 2;

if (1)                   // Always true
        a = a * 2;

if (n > 0) ;             // Wrong
        a = a * 2;
else ;
        a = a / 2;

if (n > 0)       // Nested if-else
        if (a > b)
                c = c + 1;
        else
                c = c – 1;
```

Branch statements

- if-else statement:

  Nested if-else, logic condition on the same variable:

```
if (gpa >= 8.5)
    rank = "Excellent";
else
    if (gpa >= 7.0)
        rank = "Good";
    else
        if (gpa >= 5.0)
            rank = "Fair";
        else
            rank = "Failed";
```

```
if (gpa >= 8.5)
    rank = "Excellent";
else if (gpa >= 7.0)
    rank = "Good";
else if (gpa >= 5.0)
    rank = "Fair";
else
    rank = "Failed";
```

Branch statements

- switch-case statement:
  - Syntax:

```
switch (<expression>)
{
    [case <value 1>:
        <Statement 1>;
        break;
    case <value 2>:
        <Statement 2>;
        break;
    ....]
    [default:
        <Statement N>;]
}
```

```
// Equivalent if-else statement.
if (<expression> == <value 1>)
    <Statement 1>;
else if (<expression> == <value 2>)
    <Statement 2>;
...
else
    <Statement N>;
```

Branch statements

- switch-case statement:

```
switch (day_of_week)
{
        case 1:
                printf("Sunday"); break;
        case 2:
                printf("Monday"); break;
        case 3:
                printf("Tuesday"); break;
        case 4:
                printf("Wednesday"); break;
        case 5:
                printf("Thursday"); break;
        case 6:
                printf("Friday"); break;
        case 7:
                printf("Saturday"); break;
}
```

Branch statements

- switch-case statement:
  - Notes:

    Expression encloses ( ).

    Value of **case**:

    ➔ Single value.

    ➔ Not range value.

    Statement **break**:

    ➔ Exit the **case**.

    ➔ Ignore to combine **case**.

```
switch a + b      // false
{
        ...
}

switch (a + b)
{
        case > 5:     // false
                ...
}
```

Branch statements

- switch-case statement:

```
switch (day_of_week)
{
    case 2:
    case 3:
    case 4:
    case 5:
    case 6:
        printf("Working day"); break;
    case 1:
    case 7:
        printf("Weekend"); break;
    default:
        printf("Invalid day of week");
}
```

Contents

- Branch statements.
- **Loop statements.**

Loop statements

- Consider the program:
  - Print integers in range [1..10].

    Write 10 output statements.
  - Print integers in range [1..100]?

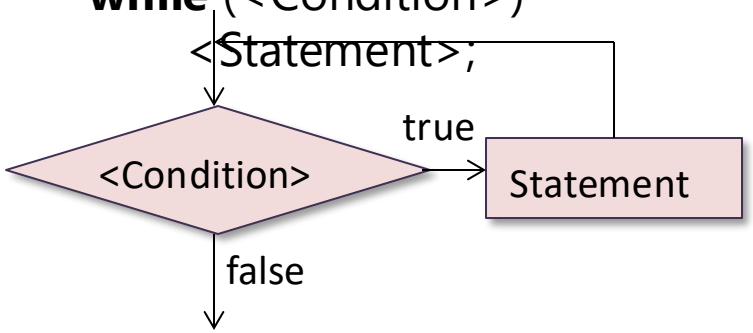
    Write 100 output statements?!

    ➔ Use loop statement.

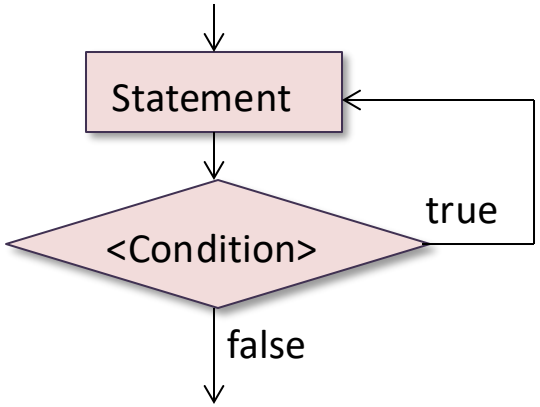
Loop statements

- while and do-while statements:
  - Syntax:

```
// while statement
while (<Condition>)
        <Statement>;

// Equivalent while for do-while
        <Statement>;
while (<Condition>)
        <Statement>;
```




```
// do-while statement
do
{
        <Statement>;
} while (<Condition>);
```



Loop statements

- while and do-while statements:

Examples:

```
// while statement.
printf("Enter n = ");
scanf("%d", &n);

i = 1;
while (i <= n)
{
    printf("%d", i);
    i++;
}
```

```
// do-while statement.
printf("Enter n = ");
scanf("%d", &n);

i = 1;
do
{
    printf("%d", i);
    i++;
} while (i <= n);
```

Loop statements

- while and do-while statements:
  - Notes:

    Loop condition encloses ( ).

    Loop often includes:

    ➔ Step 1: Initialize counter.

    ➔ Step 2: Check condition.

    ➔ Step 3: Execute a loop.

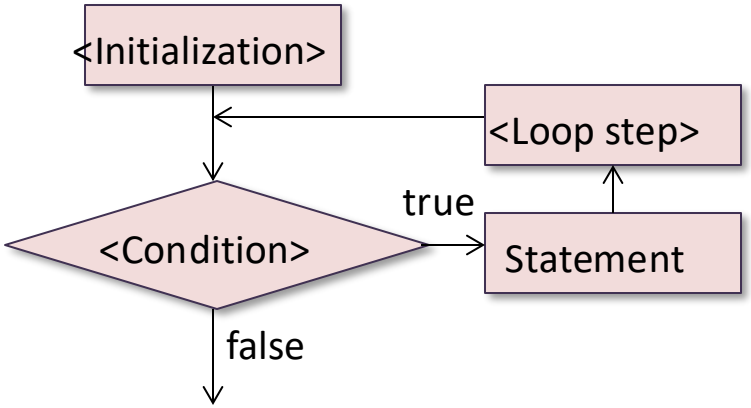
    ➔ Step 4: Update counter.

```
while  n > 0        // false
{
        ...
}

k = 0;                  // Step 1
while (k < n)           // Step 2
{
        S = S * k;   // Step 3
        k = k + 1;   // Step 4
}
```

Loop statements

- for statement:
  - Syntax:

```
for ([<Initialization>] ; [<Condition>] ; [<Loop step>])
      <Statement>;

// Equivalent while.
<Initialization>;
while (<Condition>)
{
      <Statement>;
      <Loop step>;
}
```



Loop statements

- for statement:

Examples:

```
printf("Enter n = ");
scanf("%d", &n);

// Full for.
for (i = 1; i <= n; i++)
    printf("%d", i);
```

```
printf("Enter n = ");
scanf("%d", &n);

// No initialization.
i = 1;
for ( ; i <= n; i++)
    printf("%d", i);
```

```
printf("Enter n = ");
scanf("%d", &n);

// No loop step.
i = 1;
for ( ; i <= n; )
{
    printf("%d", i);
    i++;
}
```

Loop statements

- break and continue statements:
  - break:

    Exit the current loop.

    Use with **if-else** for exit condition.
  - continue:

    Skip the current loop one time.

    Use with **if-else** for skip condition.

```
printf("Enter n = ");
scanf("%d", &n);

for (i = 1;  ; i++)
{
        if (i > n)
            break;
        if (i % 2 == 0)
            continue;

        printf("%d", i);
}
```

Summary

- Branch statements:
  - if-else: two branches.
  - switch-case: multiple branches.
- Loop statements:
  - while: condition is checked before the loop.
  - do-while: condition is checked after the loop.
  - for:

    Initialize counter.
    Check condition.
    Execute loop.
    Update counter.
  - break, continue: use with if-else in a loop.

Practice

- Practice 3.1:

Write C/C++ program to simulate a calculator as follow:

- Enter two integers.
- Enter an operator (+, -, *, /, %).
- Perform the operator on two integers and print result.

Notes: flush the standard input stream after each input.

- C: fgets, or while getchar, C++: cin.getline, or cin.ignore.

Input format:

Enter two integers = 7 5

*Enter an opertor (+, -, *, /, %) = +*

Output format (no error):

Result = 12

Output format (divided-by-zero error):

Error: divided by zero.

Practice

- Practice 3.2:

  Write C/C++ program to solve quadratic equation: $ax^2 + bx + c = 0$.

  Input format:

  Enter coefficients a, b, c = 2 -5 3

  Output format (2 solutions):

  Solution 1 = 1

  Solution 2 = 1.5

  Output format (1 solution):

  Solution 1 = <result>

  Output format (no solution):

  No solution!

Practice

- Practice 3.3:

Write C/C++ program to count the number of days in a month:

- Enter month and year.
- Count the number of days in the month and print result.

Input format:

Enter month and year = 6 2012

Output format:

*Month **6** in year **2012** has **30** days.*

Practice

- Practice 3.4:

Write C/C++ program as follow:

- Enter a positive integer N.
- Compute and print results:

a) N! = 1 * 2 * … * N.

b) ln(2) = 1 - 1/2 + 1/3 - … +/- 1/N.

c) PI = 4 (1 – 1/3 + 1/5 – … +/- 1/(2*N + 1)).

d) S = a_1 + a_2 + … a_k ({ a_i } are all square numbers <= N).

Output format:

N! = <a result>

ln(2) = <b result>

PI = <c result>

S = <d result>

Practice

- Practice 3.5:

Write C/C++ program to find all numbers satisfying:

- A 3-digit positive integer.
- Tens digit = Hundreds digit + Ones digit.

Output format:

1: <the 1st satisfied number>

2: <the 2nd satisfied number>

…

N: <the Nth satisfied number>

There are N satisfied numbers.

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&
A